

B. Tech Degree III Semester Examination, November 2008**ME 303 THERMAL ENGINEERING I**
(1999 Scheme)

Time : 3 Hours

Maximum Marks : 100

- I. (a) Give the two statements of second law of thermodynamics. (5)
 (b) Define Entropy. Derive the equation for finding the change in entropy for an ideal gas. (5)
 (c) Air is throttled from 900 KPa and 27°C to 90 KPa in a steady flow adiabatic process. If changes in kinetic and potential energies are negligible, determine the change in entropy per Kg of air flow $C_v = 1.005 \text{ KJ/Kg K}$ $R = 0.287 \text{ KJ/Kg K}$. (10)
- OR**
- II. (a) What process constitute the carnot cycle? (5)
 (b) A cyclic machine receives 325 KJ from a 1000 K energy reservoir. It rejects 125 KJ to a 400 K energy reservoir and the cycle produces 200 KJ of work as out put. Is this cycle reversible or irreversible? (5)
 (c) State and prove Classius inequality. (10)
- III. (a) Explain 'knocking' and detonation. (5)
 (b) What is the concept of valve timing diagram? What is meant by valve overlap? (5)
 (c) A 4 – cylinder, 4 stroke petrol engine is designed to develop 36 KW indicated power at a speed of 3000 rpm. The compression ratio is 5. The law of compression and expansion is $PV^{1.3} = \text{Constant}$. The pressure and temperature at the beginning of compression stroke are 1 bar and 50°C. The maximum pressure of the cycle is limited to 30 bar. Calculate the diameter and stroke of each cylinder assuming all cylinders have equal dimensions. Take diagram factor as 0.8 and $L/d = 1.5$. (10)
- OR**
- IV. (a) Explain stages of combustion in diesel engines. Define volumetric efficiency applied to 1 C engines. (10)
 (b) A 4 – cylinder, 4 stroke oil engine 10 cm in diameter and 15 cm in stroke develop a torque of 1850 Nm at 2000 rpm. The oil consumption is 15 litres/hr. The specific gravity of the oil is 0.83 and cv of oil is 42,500 KJ/Kg. If the indicated mean effective pressure taken from the indicated diagram is 6.7 bar, find –
 (i) Mechanical efficiency
 (ii) Break thermal efficiency
 (iii) Break mean effective pressure
 (iv) Specific fuel consumption. (10)
- V. (a) Derive an expression for shaft work of a single stage compressor (i) neglecting clearance (ii) with clearance. (10)
 (b) A single stage, single acting air compressor is belt driven from an electric motor at 400 rpm. The diameter and stroke of the cylinder are 15.25 cm and 17.80 cm respectively. The air is compressed from 1.05 bar to 7.35 bar and law of compression is $PV^{1.3} = \text{Constant}$. Find out the power of the motor if the transmission and mechanical efficiencies are 96% and 90% respectively. Neglect the clearance. (10)

OR*(Turn Over)*

- VI. (a) Compare rotating and reciprocating compressors. (5)
 (b) Distinguish between rotary displacement blowers and turbo blowers. (5)
 (c) A two stage centrifugal compressor delivers 500 m^3 of free air per minute. The suction condition of the air is 1 bar and 15°C . The compression ratio and is entropic efficiency of each stage are 1.25 and 80%. Find the isentropic efficiency of entire compression process. (10)
- VII. (a) Derive the general heat conduction equation in Cartesian coordinate. (10)
 (b) A furnace wall is made of refractory bricks 350 mm thick. The inner and outer surfaces of the wall have temperature of 900°C and 100°C . Find the heat loss per sq m per hour. Assume thermal conductivity of refractory brick as 4.5 W/m K . If the outside temperature becomes 50°C the furnace wall is covered with insulating bricks of 200 mm thickness. Find the reduction in heat loss. Take thermal conductivity of insulating brick as 0.5 W/m K . (10)
- OR**
- VIII. (a) State and prove Kirchoff's law of radiation. (10)
 (b) A body at 1000°C in black surroundings at 500°C has an emissivity of 0.42 at 1000°C and an emissivity of 0.72 at 500°C . Calculate the rate of heat loss by radiation per m^2 when the body is not grey. (10)
- IX. (a) Differentiate free and forced convection. (5)
 (b) Explain the physical meaning of the following dimensionless number : (5)
 (i) Nusselt number (ii) Prandtl number.
 (c) In a single stage cross flow heat exchanger both fluid unmixed, the inlet temperature of hot and cold fluid are 60°C and 35°C respectively. Calculate the effectiveness and LMTD of the heat exchanger, if heat capacity of hot fluid is less than cold fluid. (10)
- OR**
- X. (a) What is LMTD? Derive the expression for the LMTD for a counter flow heat exchanger. (10)
 (b) A heat exchanger has 17.5 m^2 area available for heat transfer. It is used for cooling oil at 200°C by using water available at 20°C . The mass flow and specific heat of oil are 10000 Kg/hr and 1.9 KJ/Kg K and the mass flow and specific heat of water are 3000 Kg/hr and 4.187 KJ/Kg.K . If the overall heat transfer coefficient is $300 \text{ W/m}^2 \text{ K}$, estimate the outlet temperature of oil and water for parallel flow by using LMTD method. (10)

